Yuxuan Zhu

AI Agent Environment and Evaluation & LLM Reinforcement Post-Training & AI and Statistics for Analytics
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RESEARCH INTERESTS

- Rigorous environments and evaluation for AI agents in critical domains (e.g., cybersecurity, data engineering).
- Reinforcement Learning with Verifiable Rewards (RLVR) for LLMs
- Efficient, theoretically sound systems for AI-powered data analytics.

EDUCATION

University of Illinois Urbana-Champaign August 2023 — present, exp. May 2028
Ph.D., Computer Science
Advisor: Daniel Kang

University of Illinois Urbana-Champaign August 2023 — December 2025
Master of Science, Computer Science

University of Michigan August 2021 — April 2023
Bachelor of Science and Engineering, Computer Science

Shanghai Jiao Tong University September 2019 — August 2023
Bachelor of Science, Electrical and Computer Engineering

RESEARCH EXPERIENCE

Kang Lab, University of Illinois Urbana-Champaign August 2023 — present

- Built **ABC**, the first checklist and assessment framework for the rigorous AI agent evaluation; NeurIPS 2025; first place at Berkeley AgentX Summit (Benchmark & Evaluation Track).
- Built **HPTSA**, a multi-agent framework that autonomously exploits zero-day web vulnerabilities; media mention by Tech Radar, Axios, The Register, New Scientist, and Fortune.
- Built **CVE-Bench**, the first benchmark evaluating AI agents' ability to exploit 40 real-world severe CVEs; adopted by US AI Safety Institute; ICML spotlight; SafeBench 2025 winner; second place at Berkeley AgentX Summit (AI Safety & Alignment Track); media mention by MIT Tech Review.
- Designed **PilotDB**, an online database-agnostic approximate query processing system with statistical guarantees; $> 5\times$ speedups on TPC-H with $\leq 5\%$ error.
- Built **UTBoost**, the first AI agent benchmark framework for software engineering tasks augmented using LLM-generated test cases; impacting 24% agents on the SWE-bench Verified Leaderboard.
- **RLVR** projects: investigating empirical generalizability of RLVR across application domains; investigating empirical and theoretical impact of reward noise in RLVR for LLMs; applying RLVR to reach expert-level text-to-SQL performance with high-quality human data (ongoing and under review).

Symbiotic Lab, University of Michigan (Advised by Fan Lai and Mosharaf Chowdhury) January 2022 — May 2023

- Built **FedTrans**, a personalized on-device distributed ML training system.

Intelligent Programming Lab, University of Michigan (Advised by Prof. Xinyu Wang) October 2021 — Apirl 2023

- Developed **SlabCity**, a whole-query optimization system leveraging program synthesis.

WORK EXPERIENCE

Data Management and Intelligence, ByteDance Inc. June 2025 — August 2025
Research Intern

SELECTED PUBLICATIONS

- **Y. Zhu**, T. Jin, C. Mo, D. Kang. "Accelerating Approximate Analytical Join Queries over Unstructured Data with Statistical Guarantees" *SIGMOD* 2026.
- **Y. Zhu**, A. Kellermann, A. Gupta, P. Li, R. Fang, R. Bindu, D. Kang. "Teams of LLM Agents Can Exploit Zero-Day Vulnerabilities" *EACL Main* 2026.
- **Y. Zhu**, T. Jin, Y. Pruksachatkun, J. Sekhon, J. Steinhardt, A. Kellermann, S. Schwettmann, M. Zaharia, I. Stoica, P. Liang, D. Kang, and others. "Establishing Best Practices for Building Rigorous Agentic Benchmarks" *NeurIPS* 2025.

- **Y. Zhu**, A. Kellermann, D. Bowman, D. Kang, and others. "CVE-Bench: A Benchmark for AI Agents' Ability to Exploit Real-World Web Application Vulnerabilities" *ICML* 2025 Spotlight.
- **Y. Zhu**, T. Jin, S. Baziotis, C. Zhang, C. Mendis, D. Kang. "Database-Agnostic Online Approximate Query Processing with A Priori Error Guarantees" *SIGMOD* 2025.
- **Y. Zhu**, D. Kang. "Efficient Approximate Query Processing with Block Sampling" *CIDR* 2025.
- **Y. Zhu**, J. Liu, F. Lai, M. Chowdhury. "FedTrans: Efficient Federated Learning via Multi-Model Transformation" *MLSys* 2024.
- **Y. Zhu**, M. Hüttner, K. Deeds. "Reproducibility Report for ACM SIGMOD 2024 Paper: Query Refinement for Diverse Top-k Selection" *SIGMOD* 2024.
- C. Hu, **Y. Zhu**, A. Kellermann, C. Biddulph, S. Waiwitlikhit, J. Benn, D. Kang. "Breaking Barriers: Do Reinforcement Post Training Gains Transfer to Unseen Domains?" *ICLR* 2026.
- S. Kapoor, B. Stroebl, P. Kirgis, N. Nadgir, Z. S Siegel, B. Wei, T. Xue, Z. Chen, F. Chen, S. Utpala, F. Ndzomga, D. Oruganty, S. Luskin, K. Liu, B. Yu, A. Arora, D. Hahm, H. Trivedi, H. Sun, J. Lee, T. Jin, Y. Mai, Y. Zhou, **Y. Zhu**, R. Bommasani, D. Kang, D. Song, P. Henderson, Y. Su, P. Liang, A. Narayanan. "Holistic Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation" *ICLR* 2026.
- M. A. Merrill, A. Shaw, N. Carlini, B. Li, H. Raj, I. Bercovich, L. Shi, J. Y. Shin, T. Walshe, E. K. Buchanan, J. Shen, G. Ye, H. Lin, J. Poulos, M. Wang, M. Nezhurina, J. Jitsev, D. Lu, O. M. Mastromichalakis, Z. Xu, Z. Chen, Y. Liu, R. Zhang, L. L. Chen, A. Kashyap, J.-L. Uslu, J. Li, J. Wu, M. Yan, S. Bian, V. Sharma, K. Sun, S. Dillmann, A. Anand, A. Lanpouthakoun, B. Koopah, C. Hu, E. Guha, G. H. S. Dreiman, J. Zhu, K. Krauth, L. Zhong, N. Muennighoff, R. Amanfu, S. Tan, S. Pimpalgaonkar, T. Aggarwal, X. Lin, X. Lan, X. Zhao, Y. Liang, Y. Wang, Z. Wang, C. Zhou, D. Heineman, H. Liu, H. Trivedi, J. Yang, J. Lin, M. Shetty, M. Yang, N. Omi, N. Raoof, S. Li, T. Y. Zhuo, W. Lin, Y. Dai, Y. Wang, W. Chai, S. Zhou, D. Wahdany, Z. She, J. Hu, Z. Dong, **Y. Zhu**, S. Cui, A. Saiyed, A. Kolbeinsson, J. Hu, C. M. Rytting, R. Marten, Y. Wang, A. Dimakis, A. Konwinski, L. Schmidt. "Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces" *ICLR* 2026
- T. Jin, Y. Choi, **Y. Zhu**, D. Kang. "Pervasive Annotation Errors Break Text-to-SQL Benchmarks and Leaderboards" *VLDB* 2026.
- T. Jin, **Y. Zhu**, D. Kang. "ELT-Bench: An End-to-End Benchmark for Evaluating AI Agents on ELT Pipelines" *VLDB* 2026.
- T. Jin, Y. Choi, **Y. Zhu**, D. Kang. "Text-to-SQL Benchmarks are Broken: An In-Depth Analysis of Annotation Errors" *CIDR* 2026.
- B. Yu, **Y. Zhu**, D. Kang, P. He. "UTBoost: Rigorous Evaluation of Coding Agents on SWE-Bench" *ACL Main* 2025.
- M. Pan, **Y. Zhu**, J. Q. Davis, R. Cogo, L. A. Agrawal, N. Arabzadeh, X. Liu, H. Mao, S. Pallerla, T. Shi, A. Xiong, A. Basile, E. Lacavalla, S. Yang, D. Arroyo, P. Castro, D. Kang, J. Gonzalez, K. Sen, D. Song, I. Stoica, M. Zaharia, M. Ellis†. "Useful Agentic AI: A Systems Outlook" *1st Workshop on Systems for Agentic AI,* 2025.
- R. Dong, J. Liu, **Y. Zhu**, C. Yan, B. Mozafari, X. Wang. "SlabCity: Whole-Query Optimization using Program Synthesis" *VLDB* 2023.
- Shijing Yuan, Jie Li, **Y. Zhu**, Chentao Wu, Yue Ding. "An Energy-efficient Computing Offloading Framework for Blockchain-enaled Video Streaming Systems" *GlobeCom* 2022.
- Shijing Yuan, Jie Li, Jinghao Liang, **Y. Zhu**, Xiang Yu, Jianping Chen, Chentao Wu. "Sharding for Blockchain based Mobile Edge Computing System: A Deep Reinforcement Learning Approach" *GlobeCom* 2021.
- M. Z Pan, N. Arabzadeh, R. Cogo, **Y. Zhu**, A. Xiong, L. A Agrawal, H. Mao, E. Shen, S. Pallerla, L. Patel, S. Liu, T. Shi, X. Liu, J. Q. Davis, E. Lacavalla, A. Basile, S. Yang, P. Castro, D. Kang, J. E. Gonzalez, K. Sen, D. Song, I. Stoica, M. Zaharia, M. Ellis. "Measuring Agents in Production" *Preprint* 2025.

SELECTED HONORS

- **Winner**, Berkeley AgentX competition, Benchmark & evaluation track. 2025
- **Winner**, Berkeley AgentX competition, AI safety & alignment track. 2025
- **Winner**, SafeBench competition. 2025
- **Winner**, Microsoft Student Hackathon - Hack for Education (AI&ML). 2021
- Undergraduate awards: Tang Junyuan Scholarship, Univ. Physics Comp. Bronze, Academic Scholarship. 2020 — 2021

Technical Skills

- Languages: Python, C/C++, SQL
- ML/LLMs: PyTorch, Transformers, VeRL, SkyRL, Tinker
- Agent / Orchestration: LangChain, OpenAI APIs, MCPs / function-calling
- Data / Systems: PostgreSQL, Spark, Ray, Docker
- Security / Evaluation: Web security (XSS, SQLi, RCE, CVEs), fuzzing, benchmark design, statistical evaluation methods

TEACHING & SERVICE

- Reviewer, ICLR 2025
- Artifact reviewer, SIGMOD 2024 — 2025
- Grader, MATH 417 Matrix Algebra, University of Michigan 2022